



# **CLEAN WATER AND SANITATION**

By: James Daniel

# Introduction

This project focuses on tackling critical challenges related to clean water and sanitation in Kenya. By addressing these challenges, the project aims to improve public health, promote sustainable water management, and ensure equitable access to basic sanitation facilities.

# SDG Alignment

- The project is aligned with Sustainable Development Goal 6: Clean Water and Sanitation.
- SDG 6 Targets:
  - Ensure availability and sustainable management of water and sanitation for all.
  - Achieve universal and equitable access to safe and affordable drinking water for all by 2030.
  - Improve water quality by reducing pollution, eliminating dumping, and minimizing the release of hazardous chemicals and materials.
  - Substantially increase water-use efficiency across all sectors.





# Problems

a. Poor sanitation infrastructure:

- 70% of the population lacks access to basic sanitation facilities, with 31% practicing open defecation.

b. Inadequate access to safe drinking water:

- Only 57% of the population has access to safe drinking water.

c. High prevalence of waterborne diseases:

- Over 4,600 cases of cholera and 14,000 cases of typhoid were reported in 2019.

d. Poor hygiene practices:

- Lack of access to basic hygiene education and facilities contributes to the spread of disease.

e. Limited financing:

- Insufficient funding hinders efforts to improve WASH infrastructure.

f. Lack of coordination among stakeholders:

- Ineffective use of resources due to duplication of efforts.

g. Climate change impacts:

- Droughts and floods exacerbate existing WASH challenges.

h. Limited government capacity:

- Challenges in implementing and enforcing WASH policies.

i. Rapid population growth:

- Puts pressure on existing WASH infrastructure and resources.

j. Cultural practices:

- Practices such as female genital mutilation and early marriage impact access to education and hygiene facilities.



# Significance

- Health Implications:

Poor WASH conditions lead to the spread of diseases, impacting community health and increasing mortality rates.

- Economic Impact:

Lack of access to clean water and sanitation hinders economic productivity and increases healthcare costs.

- Social Impact:

Women and children are disproportionately affected, with limited access to education and opportunities due to WASH challenges.

- Environmental Impact:

Inadequate sanitation infrastructure leads to environmental degradation and contamination of water sources.



# Database Design and Schema

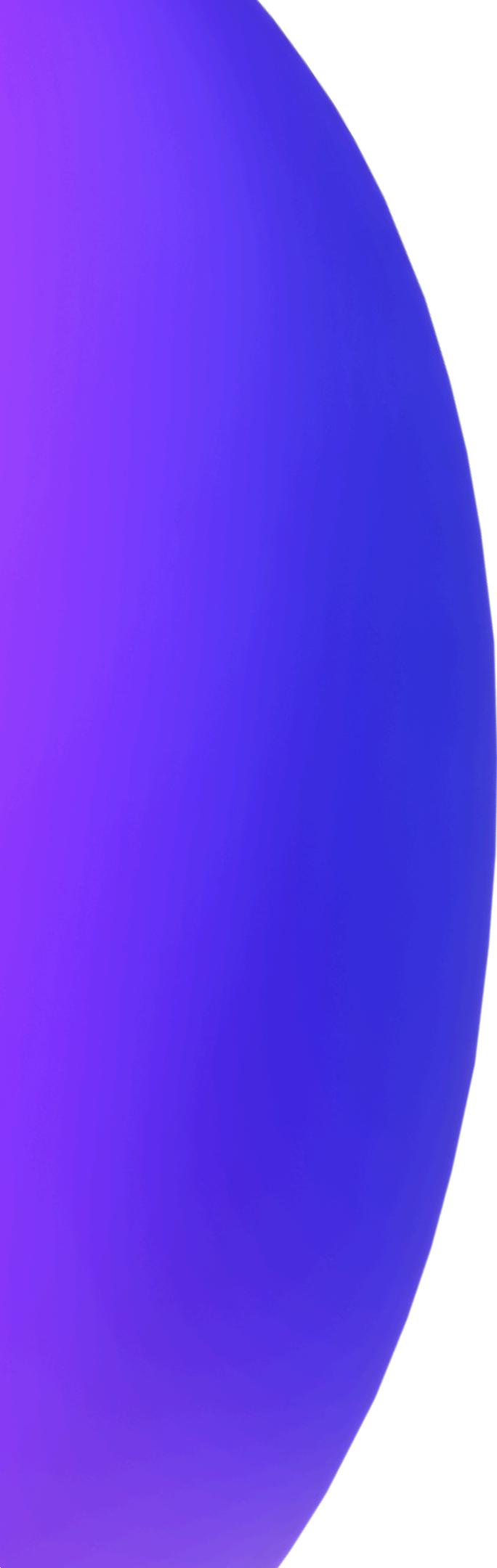
- Database Name: water\_sanitisation
- Purpose: To systematically store and manage data related to clean water, sanitation, and hygiene (WASH) issues in various communities across Kenya.
- Key Tables:
- Communities:
  - Purpose: Stores basic information about communities involved in the WASH initiatives.
- Sanitation Facilities:
  - Purpose: Tracks the types and conditions of sanitation facilities in each community.
- Water Sources:
  - Purpose: Records information on the water sources available in the community, including their quality.
- Health Records:
  - Purpose: Captures health data, particularly focusing on waterborne diseases.
- Hygiene Education Programs:
  - Purpose: Details on hygiene education programs conducted in various communities.
- WASH Funding:
  - Purpose: Tracks the funding sources and allocation for WASH-related initiatives.
- Climate Impact:
  - Purpose: Records the impact of climate events like droughts and floods on the communities.
- Stakeholders:
  - Purpose: Stores information about stakeholders involved in WASH initiatives.
- Population Growth:
  - Purpose: Tracks population growth within the communities..

# Data Analysis

- Kisumu's high population suggests a greater demand for WASH (Water, Sanitation, and Hygiene) resources.
- Reservoirs play a crucial role in providing water, particularly for regions like Kisumu and Makueni.
- The Government's significant funding contribution underscores its critical role in WASH initiatives, though private funding is also essential for comprehensive coverage.

A modern office interior with large windows and people working at desks. The scene is overlaid with a blue tint. In the center, a man in a dark shirt is standing and talking to a group of people. To his left, a woman with curly hair is standing and looking at a tablet. In the foreground, a woman with long hair is sitting at a desk, working on a laptop. Another person is sitting at a desk in the background. The office has white desks, black chairs, and large windows that look out onto a city. The text "Dashboard Demonstration" is written in a large, white, serif font across the center of the image.

# Dashboard Demonstration



## Filter

Name All ▾

Conditions All ▾

Source\_Type All ▾

Source All ▾

Disease\_Type All ▾

Population All ▾

# CLEAN WATER AND SANITATION DASHBOARD

Population

15000

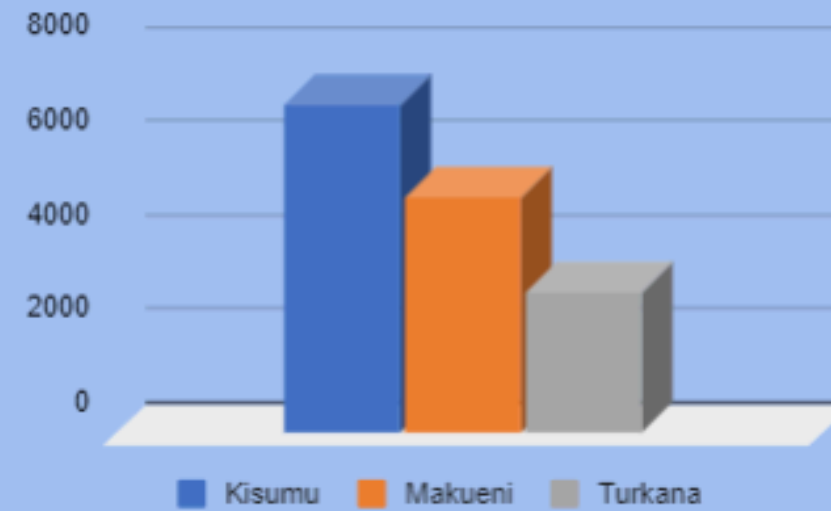
Conditions

Poor

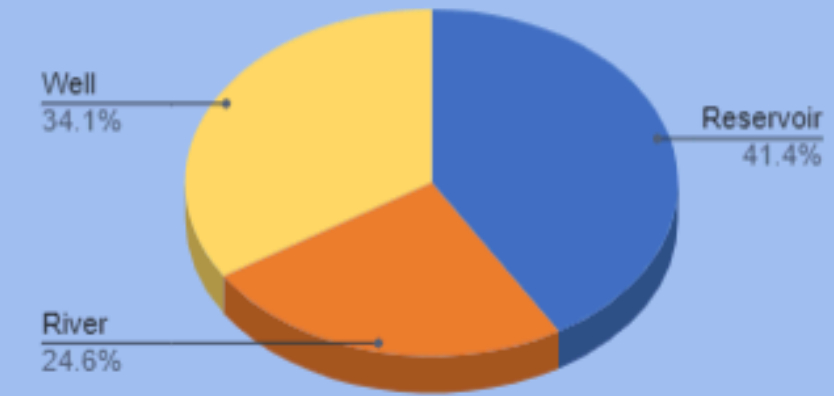
Participants

1200

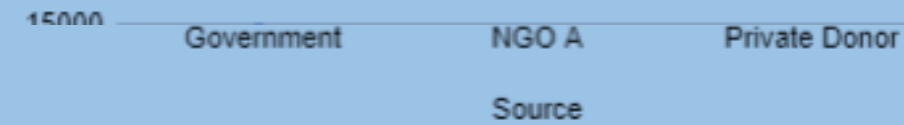
Chart population of various regions



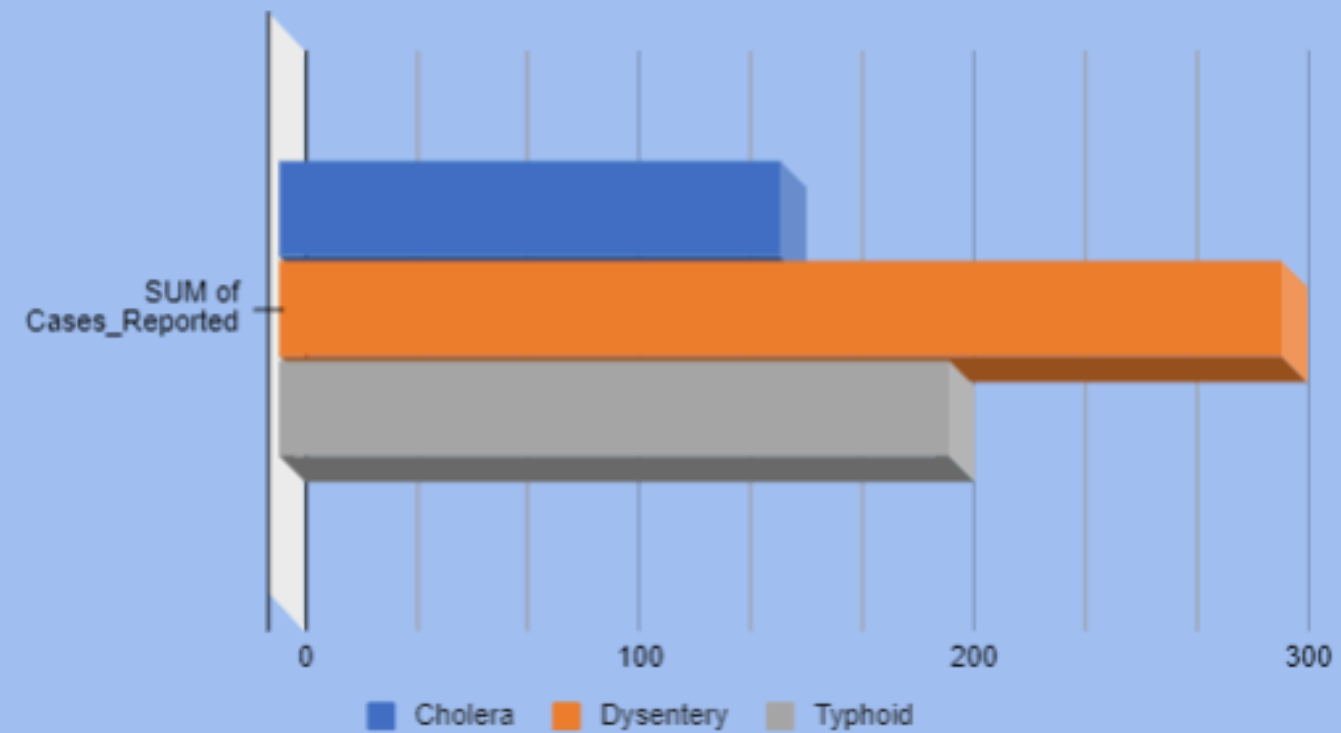
SUM of Access\_Rate



SUM of Amount



Bar Chart based on the cases reported

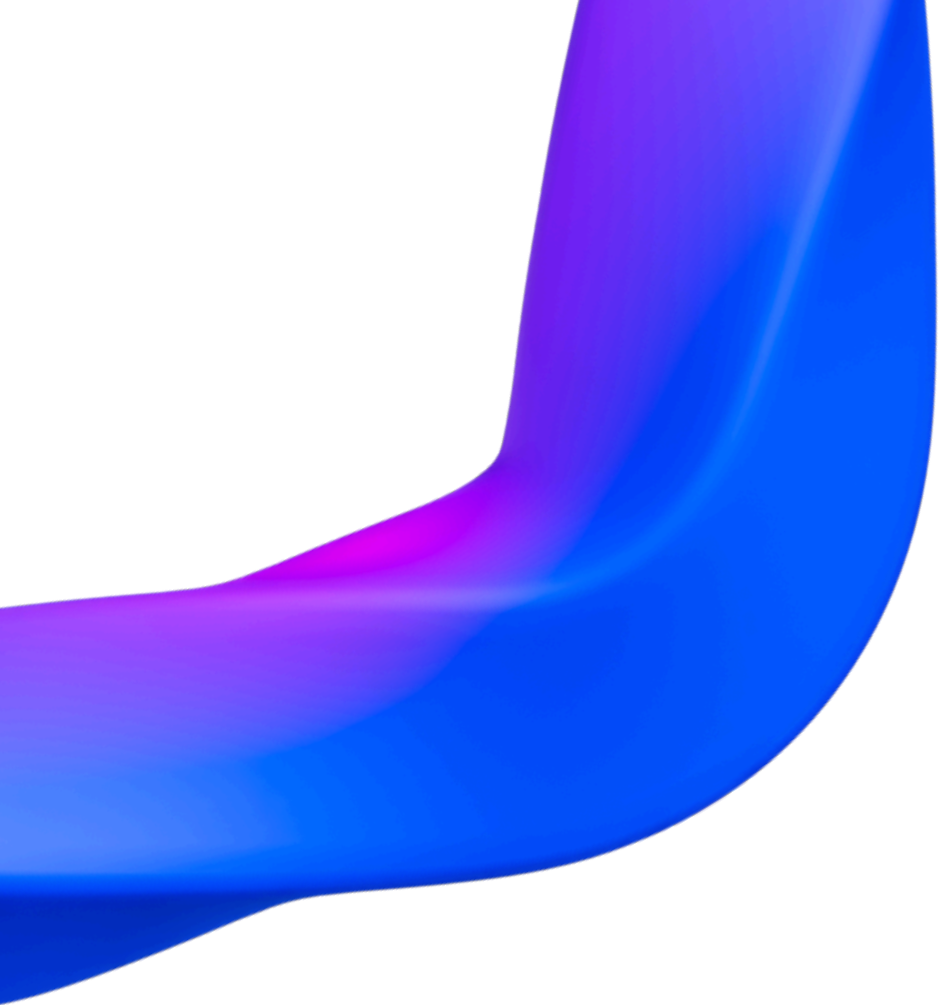


# Team



**James Daniel**

Project manager



# Thank you

## Contact Us

**James Daniel**

Project manager

✉ **Email:** [jdndirangu2020@gmail.com](mailto:jdndirangu2020@gmail.com)

🌐 **LinkedIn:** [linkedin.com/in/james-daniel](https://www.linkedin.com/in/james-daniel)

